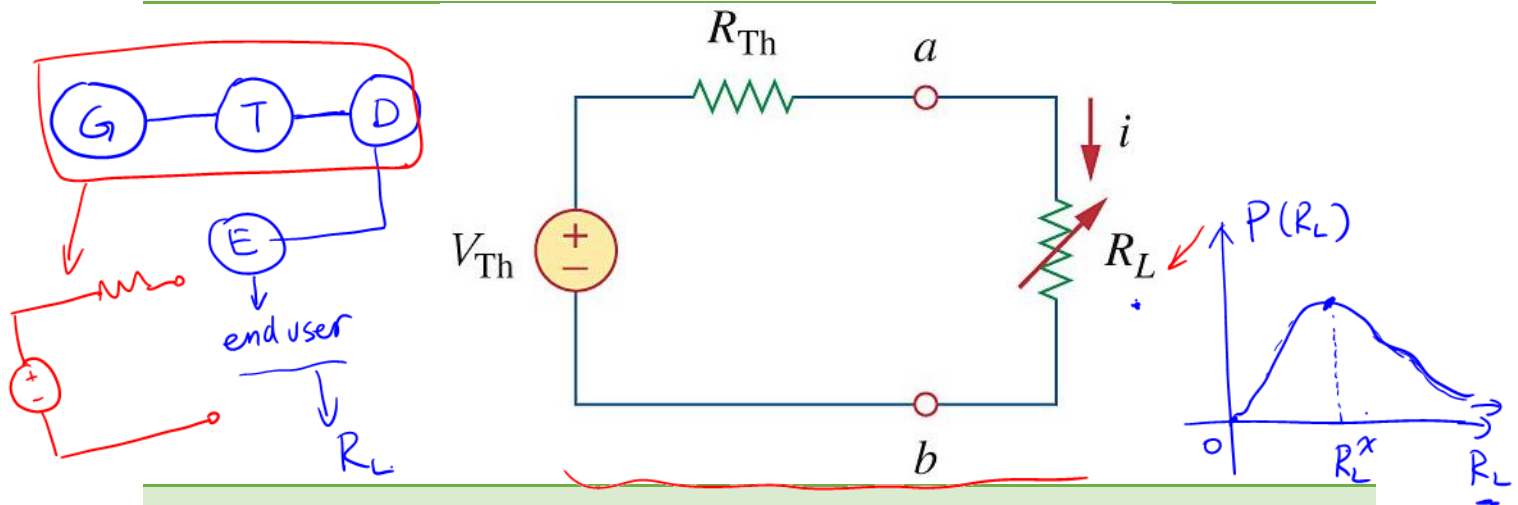


L#20: Maximum Power Transfer

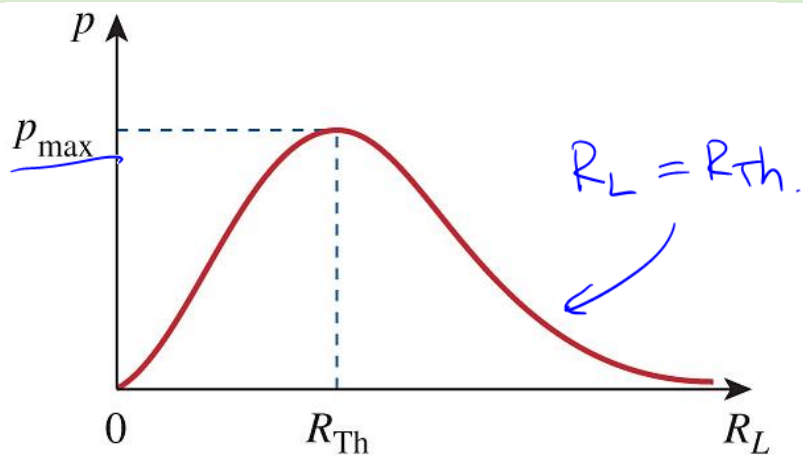


$$P(R_L) = i^2 \cdot R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L. \quad ①$$

$$R_L = 0 \Rightarrow P(R_L) = 0$$

$$R_L \rightarrow \infty \Rightarrow P = \frac{V_{Th}^2 R_L}{(R_{Th} + R_L)^2}, \quad R_L \rightarrow \infty, \quad P \rightarrow \frac{V_{Th}^2}{R_L \rightarrow \infty} \rightarrow 0$$

$$R_L \neq 0, \quad P(R_L) \neq 0.$$



The power transfer profile with different R_L

$$\underline{R_L = R_{th}} \rightarrow \underline{P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{V_{RL}^2}{R_{th}}} \quad \text{Proof?}$$

$$P = \underbrace{i^2 R_L}_{\Delta} = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L, \quad \text{let } R_L = x, \quad P = y(x).$$

$$y(x) = \frac{V_{Th}^2}{(R_{Th} + x)^2} \cdot x = \underbrace{V_{Th}^2}_{\text{constant}} \cdot (x + R_{Th})^{-2} \cdot x. \quad (1)$$

at $x = ?$, such that y is at its max value.

$$\frac{dy}{dx} = 0,$$

$$\frac{dy}{dx} = -2(x + R_{Th})^{-3} x + (x + R_{Th})^{-2} = 0 \quad (2)$$

$$(2) \quad x(x + R_{Th})^3: \quad x + R_{Th} > 0.$$

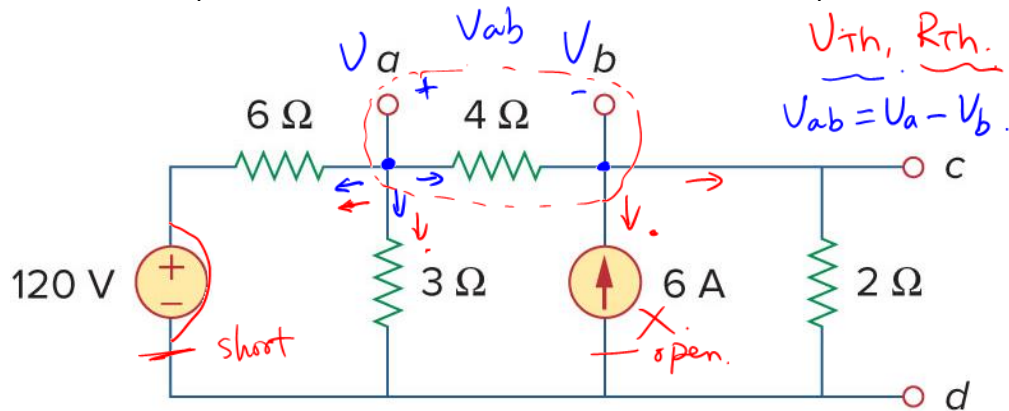
$$-2x + x + R_{Th} = 0 \Rightarrow x = R_{Th}.$$

$$\underline{x = R_L = R_{Th}}.$$

$$P_{max} = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 \cdot R_L = \frac{V_{Th}^2}{4R_{Th}^2} \cdot R_{Th} = \frac{V_{Th}^2}{4R_{Th}} \quad \checkmark$$

Example 14: Maximum Power

- Determine the value of R_L that can be connected to terminals $a-b$ that will draw maximum power from the circuit. Calculate the maximum power.



$$R_{Th} = 4 \parallel \left(\frac{6 \parallel 3 + 2}{2} \right) = 2 \Omega.$$

nodal V_a :

$$\frac{V_a - 120}{6} + \frac{V_a}{3} + \frac{V_a - V_b}{4} = 0 \quad (1)$$

$$\frac{V_a - 120}{6} + \frac{V_a}{3} - 6 + \frac{V_b}{2} = 0 \quad (2)$$

$$\left(\frac{1}{6} + \frac{1}{3} + \frac{1}{4} \right) V_a - \frac{V_b}{4} = 20$$

$$\frac{3}{4} V_a - \frac{V_b}{4} = 20 \Rightarrow 3V_a - V_b = 80 \quad (3)$$

$$\left(\frac{1}{6} + \frac{1}{3} \right) V_a + \frac{V_b}{2} = 26 \Rightarrow V_a + V_b = 52 \quad (4)$$

$$(3) + (4) \Rightarrow 4V_a = 132 \Rightarrow V_a = 33V, V_b = 19V$$

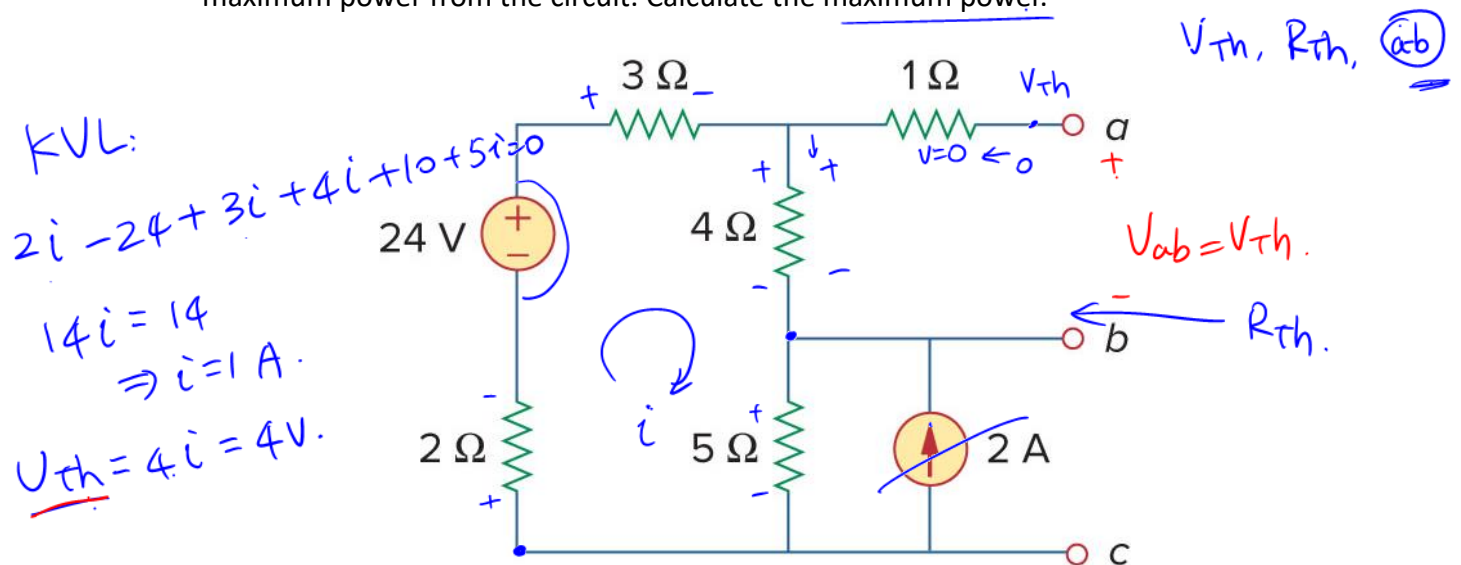
$$V_{Th} = V_{ab} = V_a - V_b = 14V.$$

$$R_L = R_{Th} = 2 \Omega.$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{14^2}{4 \cdot 2} = \text{--- W}$$

Example 15: Maximum Power

- Determine the value of R_L that can be connected to terminals $a-b$ that will draw maximum power from the circuit. Calculate the maximum power.



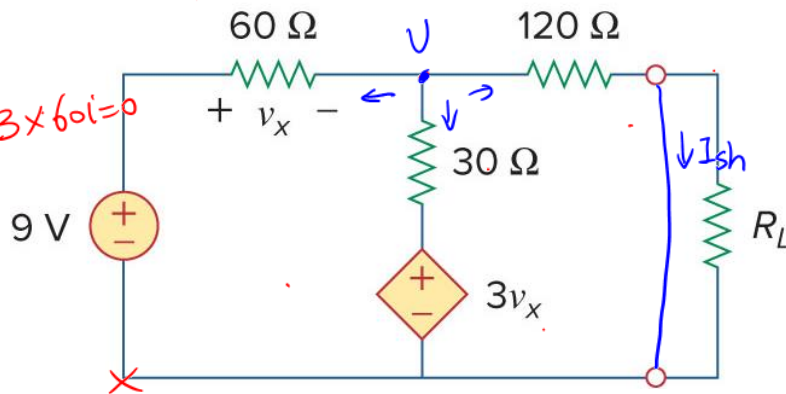
$$R_{th} = 1 + 4 \parallel (2 + 3 + 5) = 1 + \frac{4 \times 10}{4 + 10} = 1 + \frac{40}{14} = 3.86 \Omega$$

$$R_L = R_{th}, P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{4^2}{4 \times 3.86} \text{ W}$$

Example 16: Maximum Power

- Determine the value of R_L that will draw maximum power from the circuit. Calculate the maximum power.

KVL: $v_x = 60i$
 $-9 + 60i + 30i + 3 \times 60i = 0$
 $\Rightarrow 270i = 9$
 $\Rightarrow i = \frac{1}{30} \text{ A.}$



$V_{Th}, R_{Th}.$
 $V_{Th} \leftarrow \frac{V_{oc}}{I_{sh}}$
 $V_{Th} = V_{30\Omega} + 3V_x.$

$$V_{Th} = V_{30\Omega} + 3V_x = 30i + 3 \times 60i = 30 \cdot \frac{1}{30} + 3 \times 60 \cdot \frac{1}{30} = 7 \text{ V.}$$

$I_{sh} = \frac{V}{120}$, for node V.

$V_x = 9 - V. \quad (2)$

$$\frac{V-9}{60} + \frac{V-3V_x}{30} + \frac{V}{120} = 0 \quad (1)$$

$(1) \times 120: 2V - 18 + 4V - 12V_x + V = 0$

$7V - 18 - 12(9 - V) = 0$

$\Rightarrow 19V = 18 + 108 = 126 \Rightarrow V = \frac{126}{19} \text{ V.}$

$I_{sh} = \frac{V}{120} = \frac{126/19}{120} \text{ A.}, R_{Th} = \frac{V_{oc}}{I_{sh}} = \frac{7 \times 120}{126/19}$

$= \frac{7 \times 120 \times 19}{126}$
 $= 126.7 \Omega$

$R_L = R_{Th},$

$P_{max} = \frac{V_{Th}^2}{4 R_{Th}} = \frac{7^2}{4 \times 126.7} \text{ W.}$